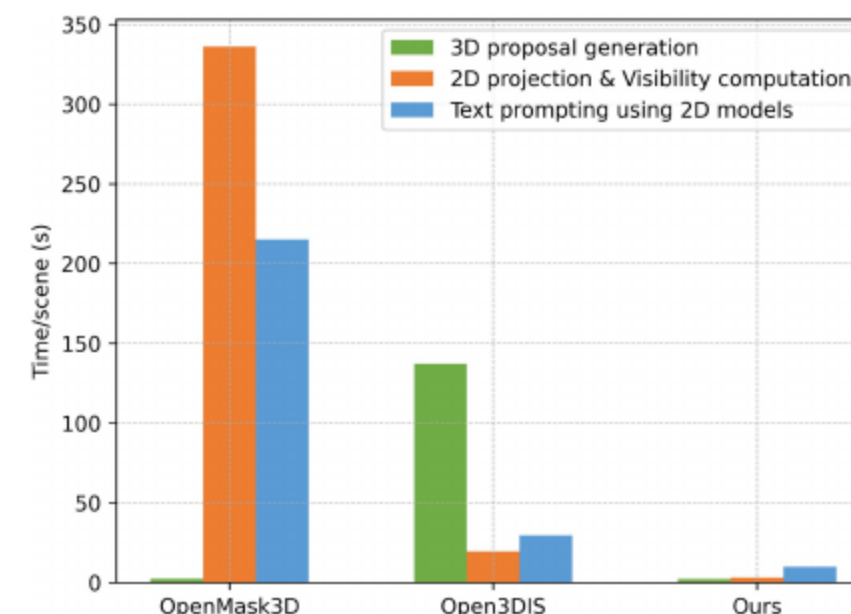


Open-YOLO 3D: Towards Fast and Accurate Open-Vocabulary 3D Instance Segmentation

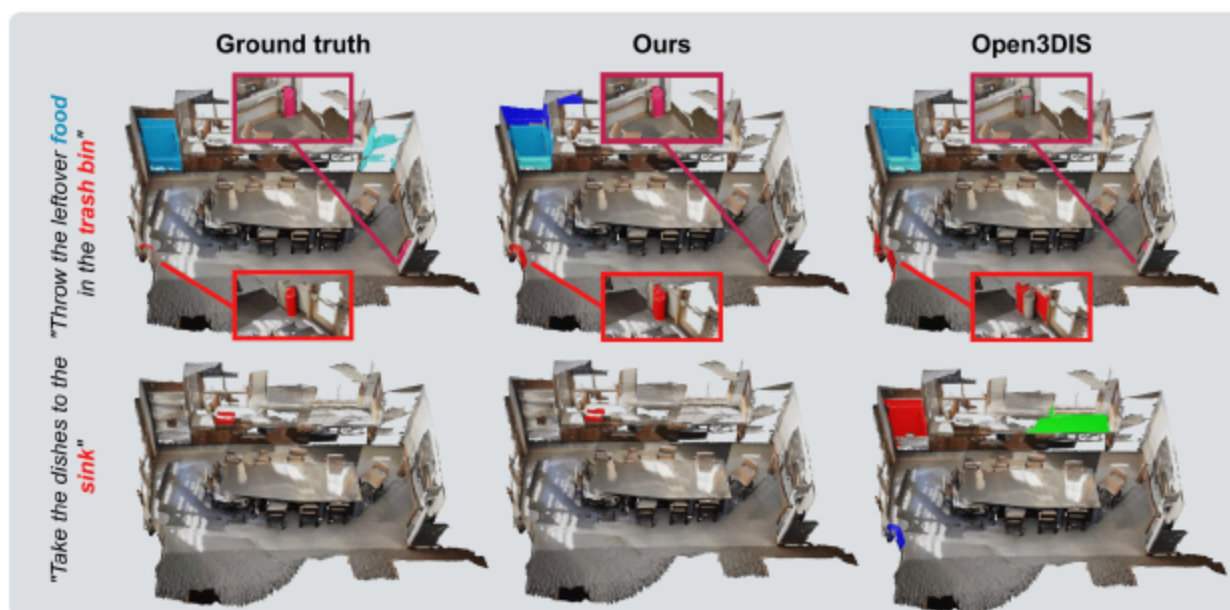
Mohamed El Amine Boudjoghra; Angela Dai;
Jean Lahoud; Hisham Cholakkal; Rao Muhammad
Anwer; Salman Khan; Fahad Shahbaz Khan

01 Motivation: Fast Open-Vocabulary 3D Segmentation



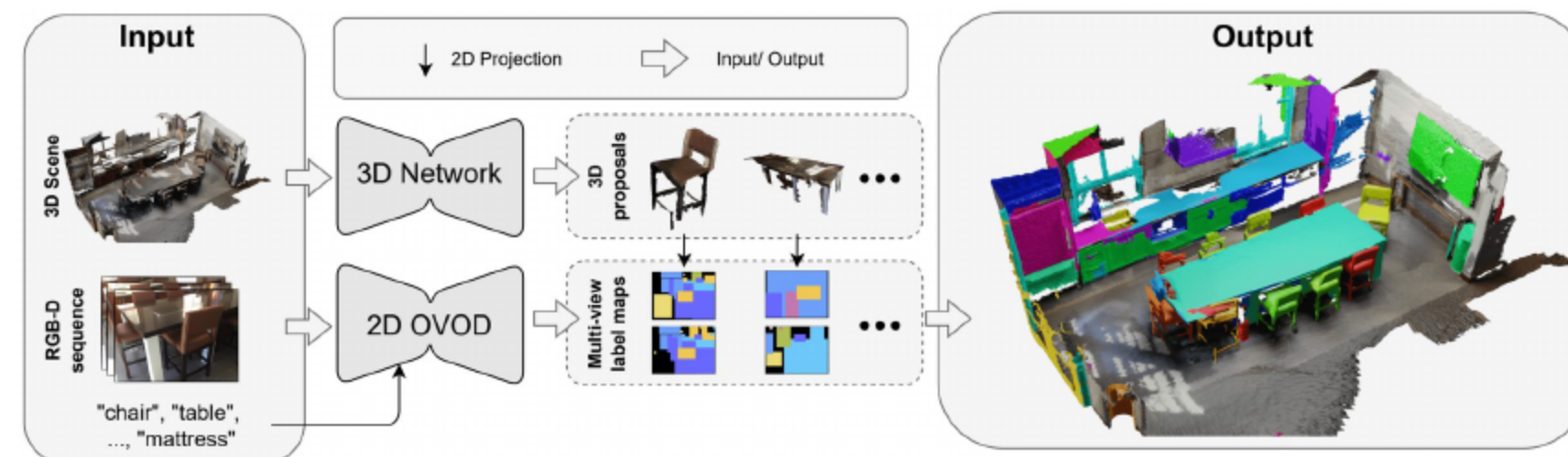
- Prior open-vocabulary 3D pipelines using SAM and CLIP are **slow**, taking about 5 to 10 minutes per scene
- Robotics and AR require low-latency, generalizable 3D labeling for novel objects in dynamic settings
- **Goal:** retain open-vocabulary generalization while achieving at least a ten times end-to-end speedup
- Heavy 2D segmentation and 2D to 3D feature lifting dominate runtime and are **computationally expensive**

02 Our Approach and Contributions: Open-YOLO 3D



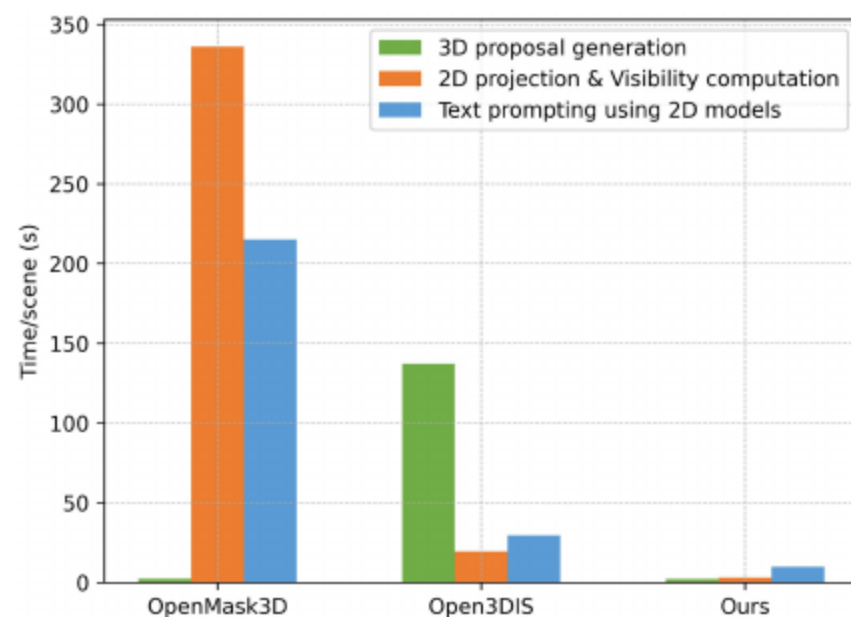
- Replace heavy 2D segmentation with open-vocabulary 2D detection and fast 3D class-agnostic masks
- Construct low granularity label maps from detector boxes; use multi-view prompt distribution for text-to-3D assignment
- confidence improves ranking
- **Contributions:** detector-based labeling, proposal scoring without per-pixel masks, and **faster** end-to-end inference
- On ScanNet200, improves mAP at 50 percent overlap with about **sixteen times** speedup over a

03 From Closed 3D to Open-Vocabulary



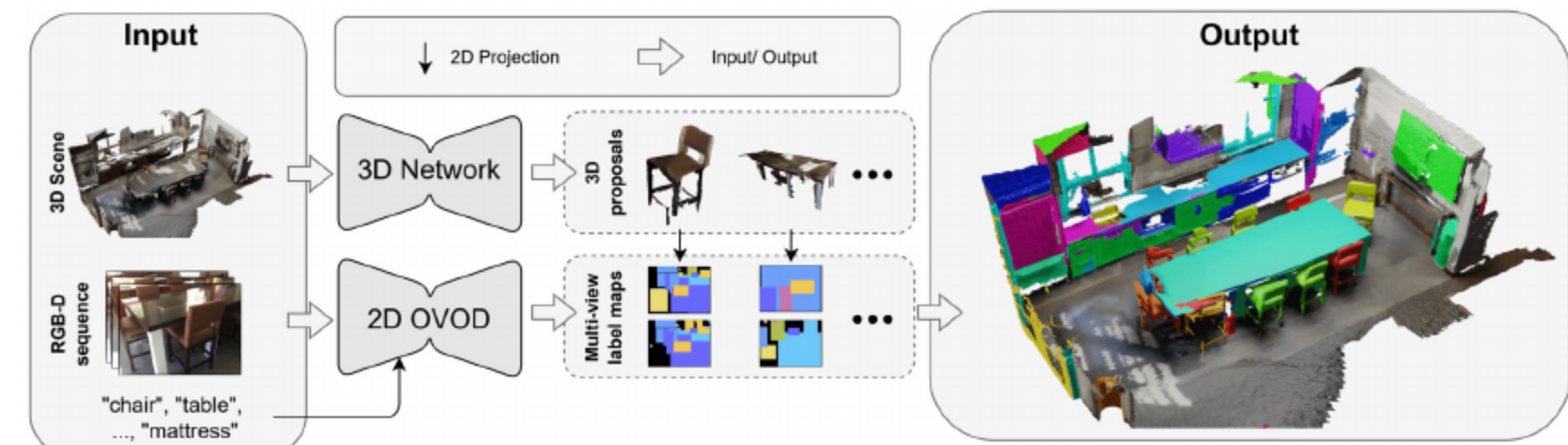
- Closed-set 3D methods span bottom-up grouping, top-down proposals, and transformer hybrids such as Mask3D
- Open-vocabulary 2D detection and segmentation use vision-language models, with modern detectors offering **strong** zero-shot capability
- Open-vocabulary 3D methods using SAM and CLIP achieve good accuracy but are **computationally expensive** at inference
- Recent 3D approaches fuse 2D masks or lift features across views, improving recall on small objects but increasing cost

04 Problem Setup and Baseline Limitations



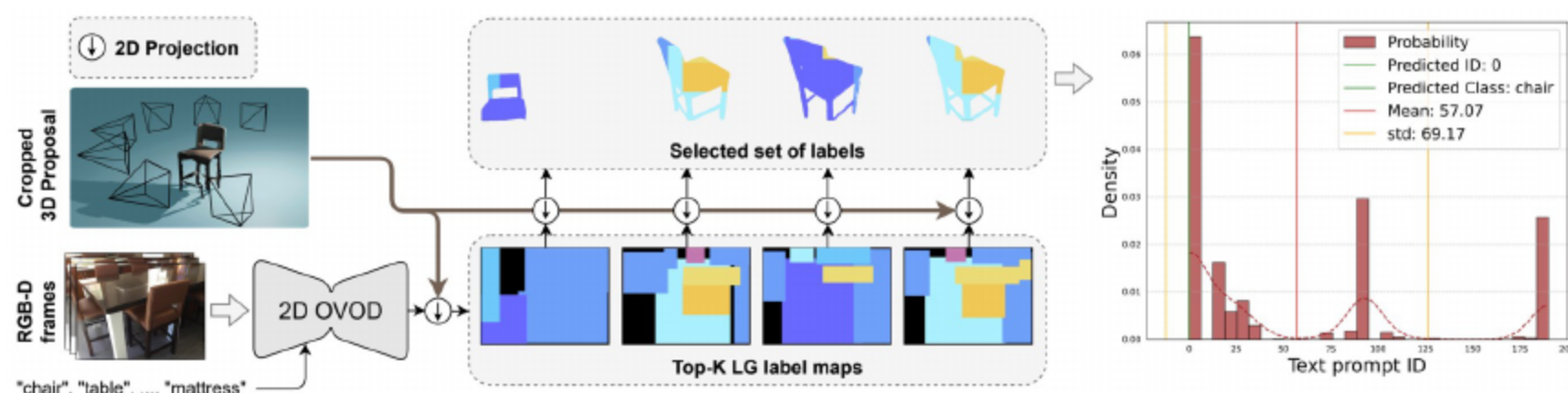
- Inputs are RGB-depth frames and camera poses; outputs are 3D instance masks with open-vocabulary labels
- Baselines dominated by 2D segmentation and 2D to 3D feature lifting, which are **slow** and resource intensive
- Key need: efficient visibility computation and multi-view labeling without dense per-pixel 2D segmentation

05 Strategy + Pipeline Overview



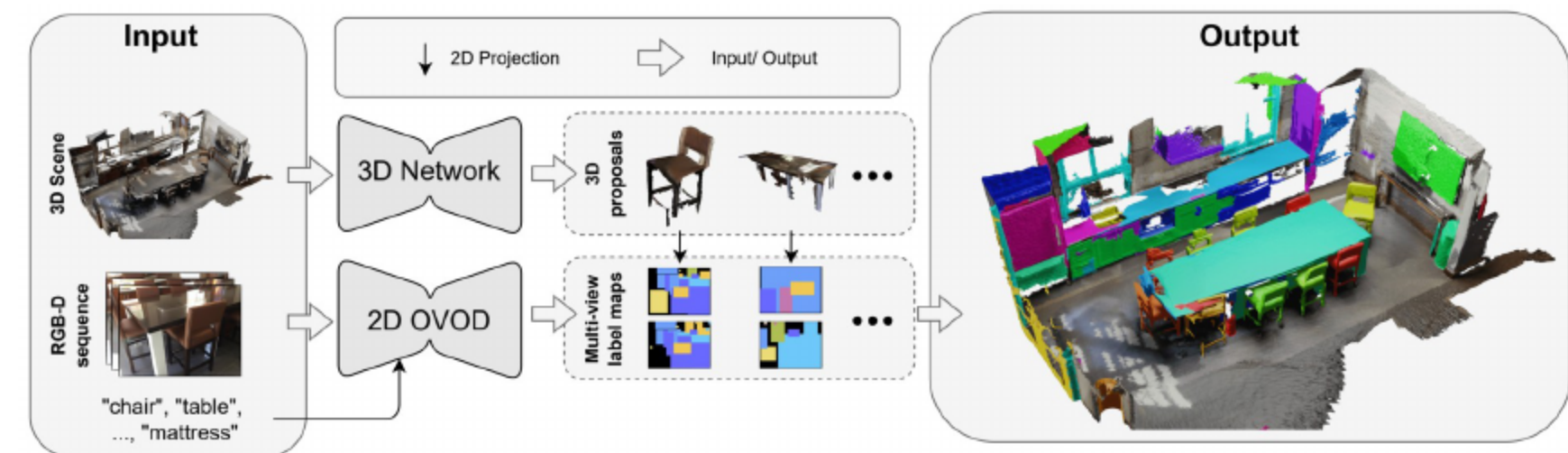
- Generate fast class-agnostic 3D proposals using a transformer-based 3D instance segmentation backbone
- Use an open-vocabulary 2D detector to produce boxes and labels that form per-frame label maps
- Compute batched visibility, select top views, and label maps
- Project all scene points once per frame batch, avoiding iterative per-mask counting that is **time-consuming**
- Outcome:** fewer 2D calls, **faster** visibility, and reliable text-to-3D matching

06 2D OV Detections to Low-Granularity Label Maps



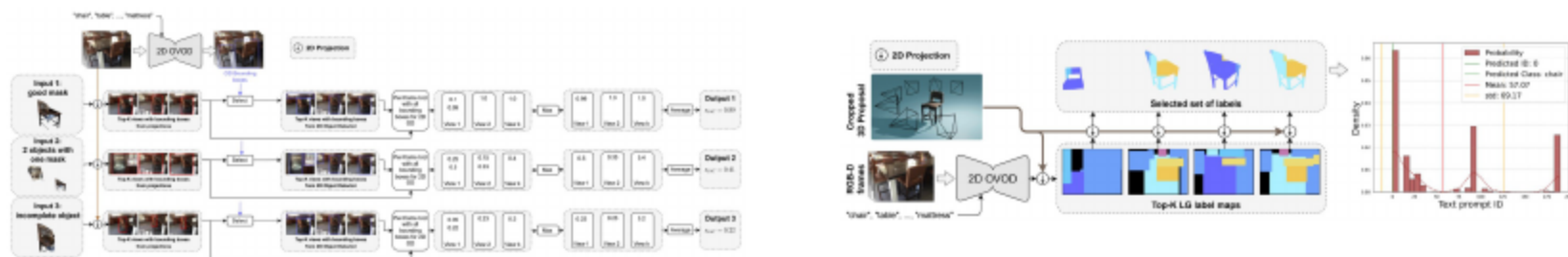
- Create label maps per frame by initializing ignore regions and filling detector box areas with predicted class identifiers
- Sort boxes by area to roughly account for occlusion ordering in overlapping regions
- These lightweight label surfaces enable **fast** multi-view voting with minimal overhead
- No** per-pixel masks are required, cutting **redundant** 2D computation

07 VAcc: Fast Visibility via Batched Projections



- Project the full point cloud into all frames in one GPU batch for efficient 2D coordinates and depths
- Compute in-frame and depth-consistent masks using tensor operations to filter occlusions
- Assemble a visibility matrix per mask and select the top visible frames per proposal
- Accelerated** visibility replaces iterative loops that are **slow**

08 MVPDist Labeling and Joint Scoring



- Multi-view prompt distribution: gather labels at projected coordinates from top views and aggregate into a class distribution
- Assign the class with the highest occurrence; compute a joint score combining mask alignment and class confidence
- Mask alignment score uses average overlap between projected mask boxes and best detector

10 SOTA Accuracy and ~16× Speedup

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	head	comm	tail	time/scene (s)
Mask3D [39] (Closed Vocab.)	×	×	26.9	36.2	41.4	39.8	21.7	17.9	13.41
SAM3D [50]	✓	×	6.1	14.2	21.3	7.0	6.2	4.6	482.60
OVIR-3D	✓	×	13.0	24.9	32.3	14.4	12.7	11.7	466.80
Open3DIS [34]	✓	×	23.7	29.4	32.8	27.8	21.2	21.8	360.12
OpenScene [35] (2D Fusion)	×	×	11.7	15.2	17.8	13.4	11.6	9.9	46.45
OpenScene [35] (3D Distill)	×	×	4.8	6.2	7.2	10.6	2.6	0.7	0.26
OpenScene [35] (2D-3D Ens.)	×	×	5.3	6.7	8.1	11.0	3.2	1.1	46.78
OpenMask3D [42]	×	✓	15.4	19.9	23.1	17.1	14.1	14.9	553.87
Open3DIS [34]	×	×	18.6	23.1	27.3	24.7	16.9	13.3	57.68
Open-YOLO 3D (Ours)	×	×	24.7	31.7	36.2	27.8	24.3	21.6	21.8

- **State-of-the-art** accuracy using only 3D proposals on ScanNet200
- About sixteen times faster than a recent open-vocabulary baseline while improving average precision at fifty percent overlap
- Eliminates SAM and CLIP lifting, enabling **efficient** end-to-end open-vocabulary labeling
- Competitive across head, common, and tail splits with lower latency

09 Setup: Datasets, Metrics, Implementation

Method	3D proposals from 2D masks	SAM for 3D mask labeling	mAP	mAP50	mAP25	head	comm	tail	time/scene (s)
Mask3D [39] (Closed Vocab.)	×	×	26.9	36.2	41.4	39.8	21.7	17.9	13.41
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- **Datasets:** ScanNet200 validation with head, common, tail splits; Replica with 48 categories
- **Metrics:** average precision at multiple overlap thresholds, including fifty and twenty five percent
- High-capacity open-vocabulary detector and class-agnostic 3D proposals with non-maximum suppression on a single A100
- Process every tenth frame on ScanNet200 and all frames on Replica for fair comparisons

11 Quantitative and Qualitative

- On Replica, higher average precision than CLIP-based baselines under 3D-only proposal settings
- Around eleven times speedup over strong CLIP pipelines, showing **efficient** generalization
- Qualitatively, improved precision on text-queried instances across challenging scenes
- Maintains accuracy while removing **redundant** mask fusion stages



12 Oracle Mask Setting: Text-to-3D Retrieval

Method	2D OV Prior	mAP	head	comm	tail
<i>Closed Vocab.</i> Mask3D [39]	None	35.5	55.2	27.2	22.2
OpenScene [35]	2D LSeg [27]	11.8	26.9	5.2	1.7
OpenScene [35]	2D OpenSeg [12]	22.9	26.2	22.0	20.2
OpenMask3D [42]	Mask wise CLIP [37]	29.1	31.1	24.0	32.9
Open3DIS [34]	Point wise CLIP [37]	30.9	33.1	25.1	35.4
Open-YOLO 3D (Ours)	MVPDist	39.6	43.3	36.8	38.5

- With ground truth masks, the multi-view prompt distribution surpasses CLIP-based retrieval on ScanNet200
- **Stronger** zero-shot behavior via multi-view voting from open-vocabulary detectors
- Consistent gains across head, common, and tail categories
- Demonstrates the benefit of detector signals without per-pixel supervision

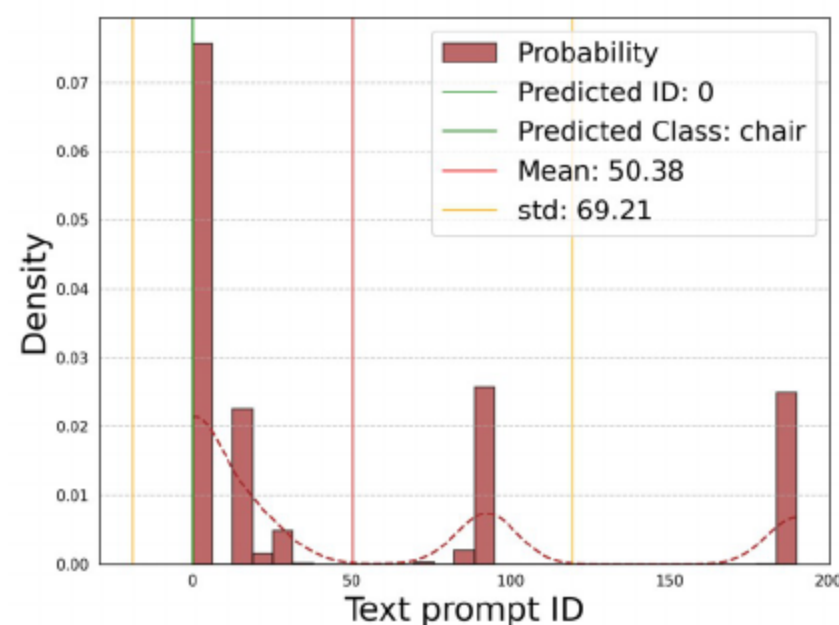
13 Detectors vs SAM/CLIP; MVPDist and VAcc

Row ID ↓	2D Prior	Labeling method	MVPDist	VAcc	mAP	time/scene (s)
R1	SAM[23]+CLIP	Clip features	×	×	33.0	675.6
R2	YoloV8+CLIP	Clip features	×	×	21.1	440.40
R3	RT-DETR[57]+CLIP	Clip features	×	×	28.4	487.95
R4	YoloWorld[7]+CLIP	Clip features	×	×	32.5	384.29
R5	YoloWorld	LG Label Maps	✓	×	46.2	376.42
R6	YoloWorld	LG Label Maps	✓	✓	46.2	17.86

- Open-vocabulary detectors can replace SAM for crop generation with **substantial** speed benefits
- Multi-view prompt distribution with label maps improves zero-shot labeling over CLIP aggregation
- The visibility accelerator reduces visibility computation time from minutes to seconds
- Detector-based crops approach SAM quality while avoiding **heavy** segmentation inference

14 Qualitative and Limits

- Top view voting improves robustness; relying on a single view often underperforms
- Low granularity maps are about five times **faster** than high granularity with similar accuracy
- Limitation: very small objects can be **missed** by 3D-only proposal generators
- High-granularity maps add **overhead** without changing predictions for most cases



THANKS!